

YOLOV8 TECHNICAL REPORT

Training Analysis & Car Bodywork Defect Detection — MyGarage

1. Executive Summary

This document provides a comprehensive technical analysis of the YOLOv8 model performance trained over 25 epochs for the **MyGarage** project. The primary objective is the precise detection of vehicle bodywork defects (dents, scratches, impacts, etc.). The model demonstrates excellent convergence and a remarkable balance between precision and recall.

0.948

PRECISION

0.853

RECALL

0.891

MAP@50

0.750

MAP@50-95

2. Learning Convergence

The evolution of loss functions and primary metrics demonstrates healthy training behavior. Both training and validation losses decrease synchronously, indicating the absence of overfitting and a strong capability of the model to generalize well on unseen test data.

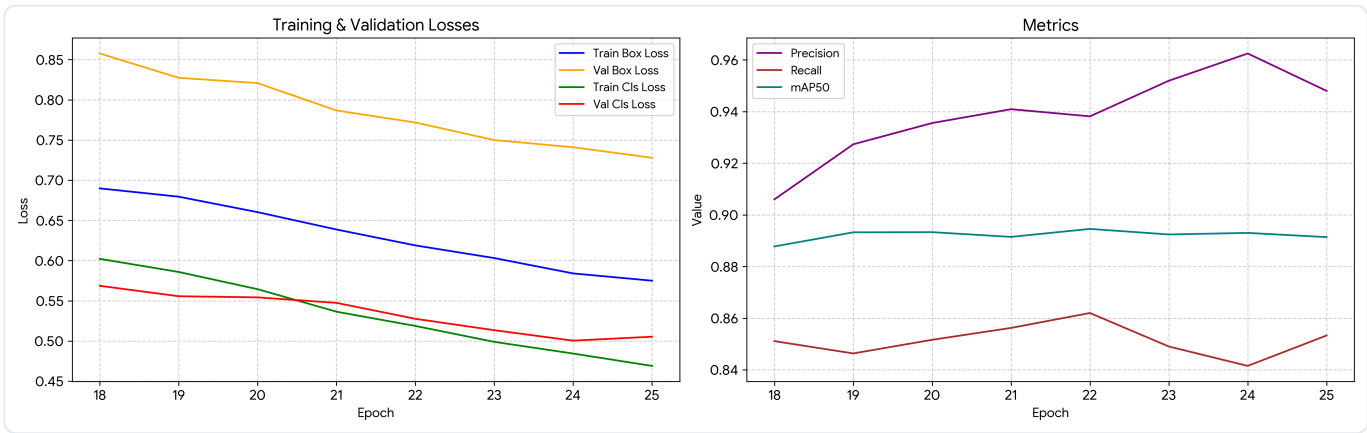


Figure 1: Evolution of training/validation losses and major metrics across epochs.

3. Evaluation via Confusion Matrix

The confusion matrix evaluates prediction distribution per class against ground truth labels. High values along the main diagonal reflect the model's capacity to precisely differentiate distinct damage categories (e.g., Bumper-dent vs. Doorouter-dent) while keeping background false positives low.

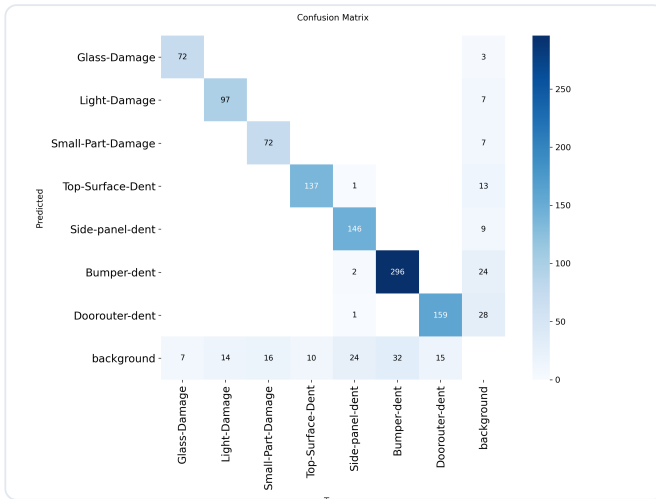


Figure 2a: Absolute confusion matrix

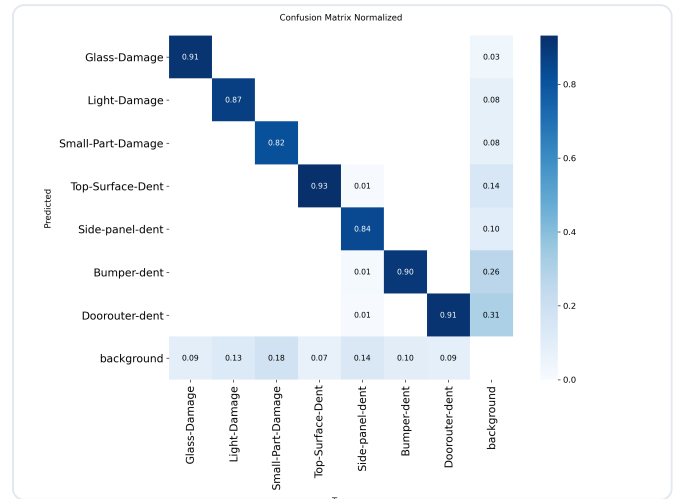


Figure 2b: Normalized confusion matrix

4. F1 and PR Performance Curves

The F1-Confidence curve establishes an optimal confidence threshold at **0.501**, achieving a peak F1-score of **0.90** across all classes combined. The Precision-Recall (PR) curve highlights detection robustness with an overall area under the curve (mAP@0.5) of 0.891.

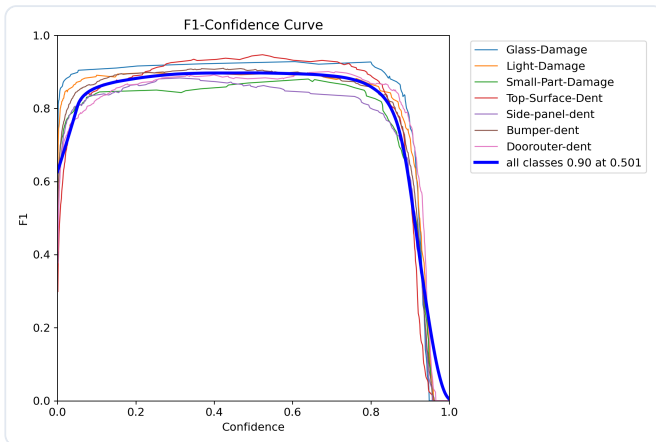


Figure 3a: F1 vs. Confidence curve

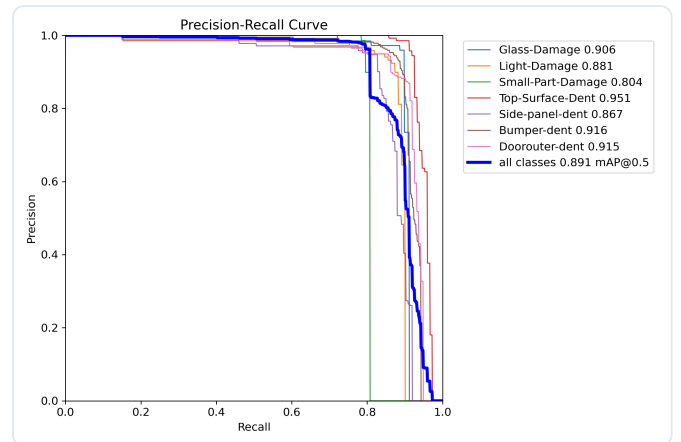


Figure 3b: Precision-Recall curve

5. Spatial Analysis and Class Distribution

The class distribution demonstrates dataset representation across defect categories. Spatial density plots (X, Y) and bounding box dimensions (Width, Height) display uniform label layouts, ensuring stable object anchor box predictions during model inference.

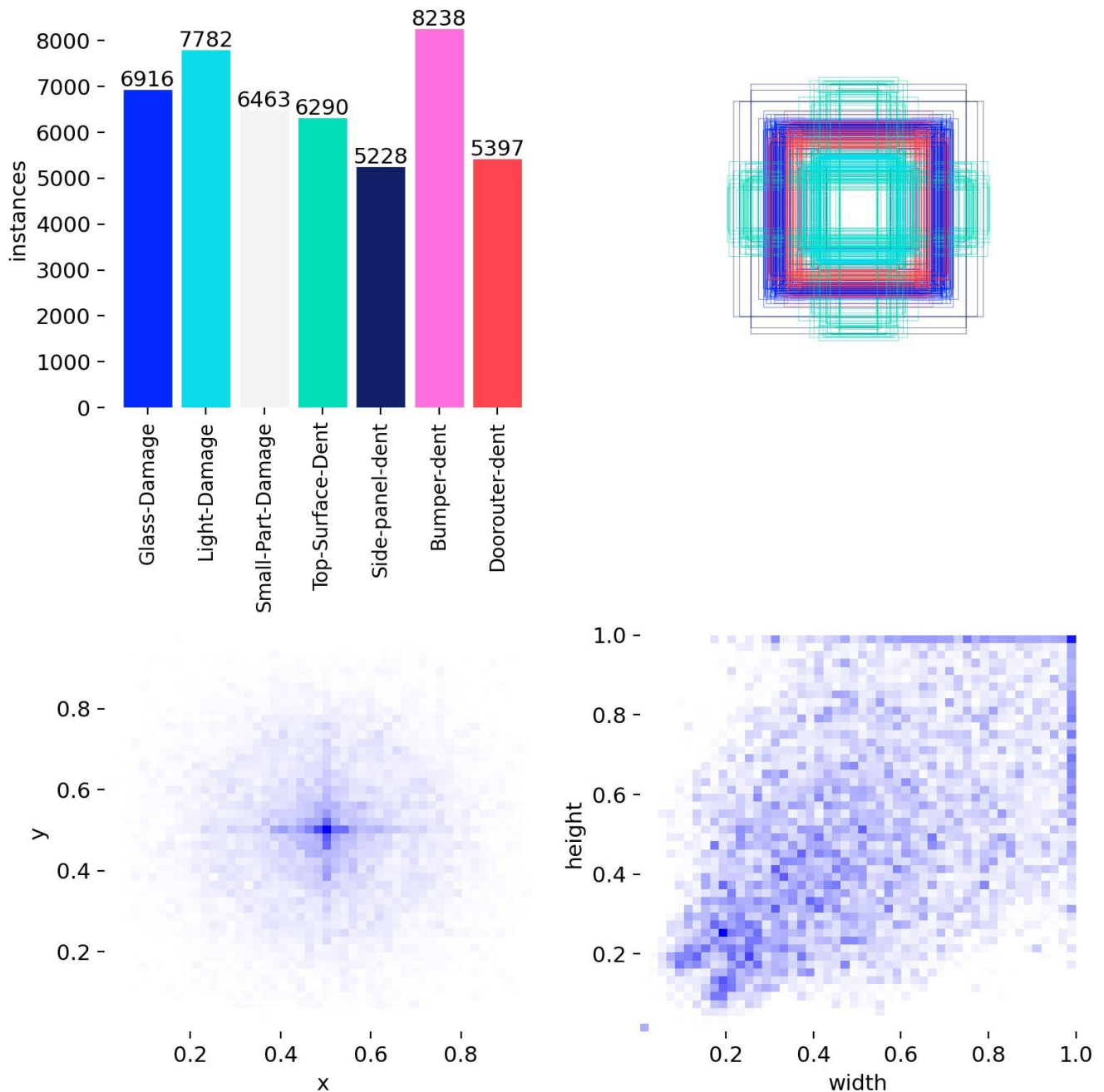


Figure 4: Core distribution and spatial feature analysis of dataset labels.

6. Resources & Download Links

Link to download the raw dataset:

[Download Raw Dataset](#)

Link to the training data after preprocessing:

[Download Preprocessed Data](#)

Link to the trained model:

[Download Trained Model Weights](#)